Linear Algebra for QC [3/3]

Tensor Products and Multi-Qubit Systems

Quantum Computing and AI

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1 qubit $\alpha|0\rangle + \beta|1\rangle$

$\mathbb{C}^2$ $\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$

1st qubit → $(\alpha|0\rangle + \beta|1\rangle)$ $\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$

second qubit → $(\alpha'|0\rangle + \beta'|1\rangle)$ $\begin{pmatrix}\alpha'\\ \beta'\end{pmatrix}$

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes \begin{pmatrix}\alpha'\\ \beta'\end{pmatrix} = \begin{pmatrix}\alpha\alpha'\\ \alpha\beta'\\ \beta\alpha'\\ \beta\beta'\end{pmatrix}$$

$|\alpha|^2+|\beta|^2=1$ ↑ $|\alpha'|^2+|\beta'|^2=1$

Q: how will we represent the joint system of the 2 qubits?

$$\underbrace{\alpha\alpha'}|00\rangle + \underbrace{\alpha\beta'}|01\rangle + \underbrace{\beta\alpha'}|10\rangle + \underbrace{\beta\beta'}|11\rangle$$

$$|\alpha\alpha'|^2 + |\alpha\beta'|^2 + |\beta\alpha'|^2 + |\beta\beta'|^2 = 1.$$

From One Qubit to Two

- One qubit: a unit vector in $\mathbb{C}^2$. $\alpha|0\rangle + \beta|1\rangle$ $|\alpha|^2 + |\beta|^2 = 1$
- Two classical bits: *one* of $00, 01, 10, 11$.
- Two qubits: a complex amplitude for *each* of $00, 01, 10, 11$, i.e. a unit vector in $\mathbb{C}^4$.

From One Qubit to Two

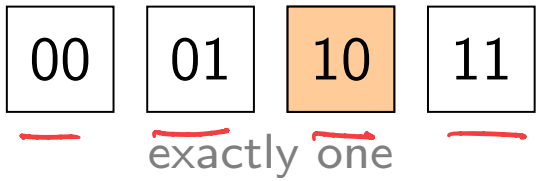
- One qubit: a unit vector in $\mathbb{C}^2$.
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Question for today

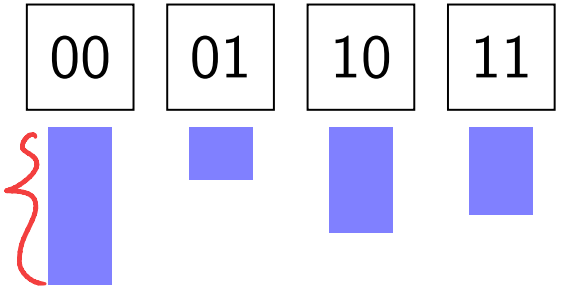
How is $\mathbb{C}^4$ built from two copies of $\mathbb{C}^2$? And what new phenomena appear?

The answer is the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2$.

2 bits



2 qubits



an amplitude on every string

Plan

$M \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$

$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|1\rangle + \beta|0\rangle$

1. Tensor product of vectors: definition, picture, rules
2. Multi-qubit states, kets, and exponential growth
3. Product states vs **entangled** states
4. Tensor product of matrices: gates on several qubits
5. Applications: creating entanglement, measuring one qubit, Bell correlations

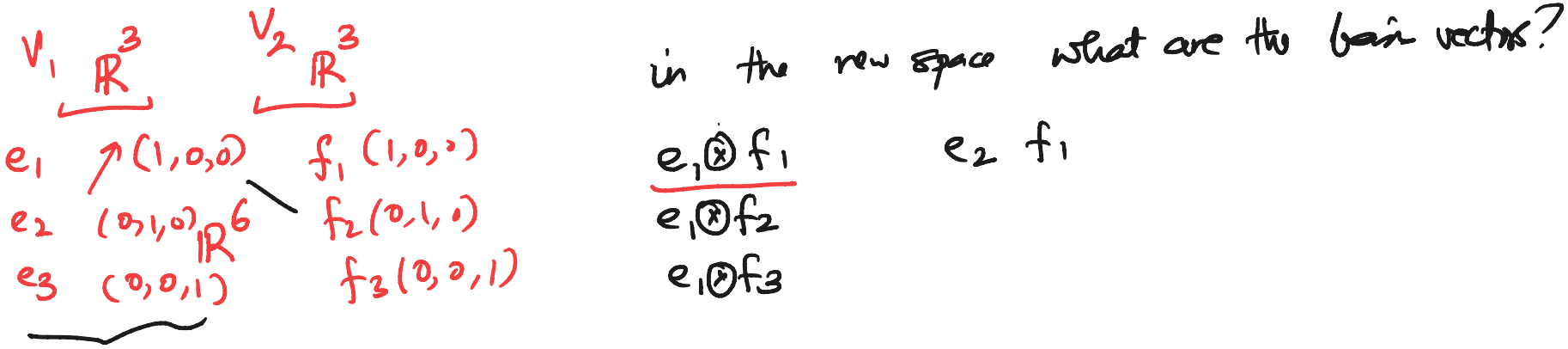
Notation: bold vectors as before. Since kets have now been introduced, examples also use $|0\rangle, |1\rangle, |+\rangle, |-\rangle$.

Tensor Product of Two Vectors

Definition (for $\mathbb{C}^2 \otimes \mathbb{C}^2$)

$$\mathbf{u} \otimes \mathbf{v} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \otimes \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 \mathbf{v} \\ u_2 \mathbf{v} \end{pmatrix} = \begin{pmatrix} u_1 v_1 \\ u_1 v_2 \\ u_2 v_1 \\ u_2 v_2 \end{pmatrix} \in \mathbb{C}^4.$$

In general, for $\mathbf{u} \in \mathbb{C}^m$, $\mathbf{v} \in \mathbb{C}^n$: stack $u_1\mathbf{v}, \ldots, u_m\mathbf{v}$ to get a vector in $\mathbb{C}^{mn}$.



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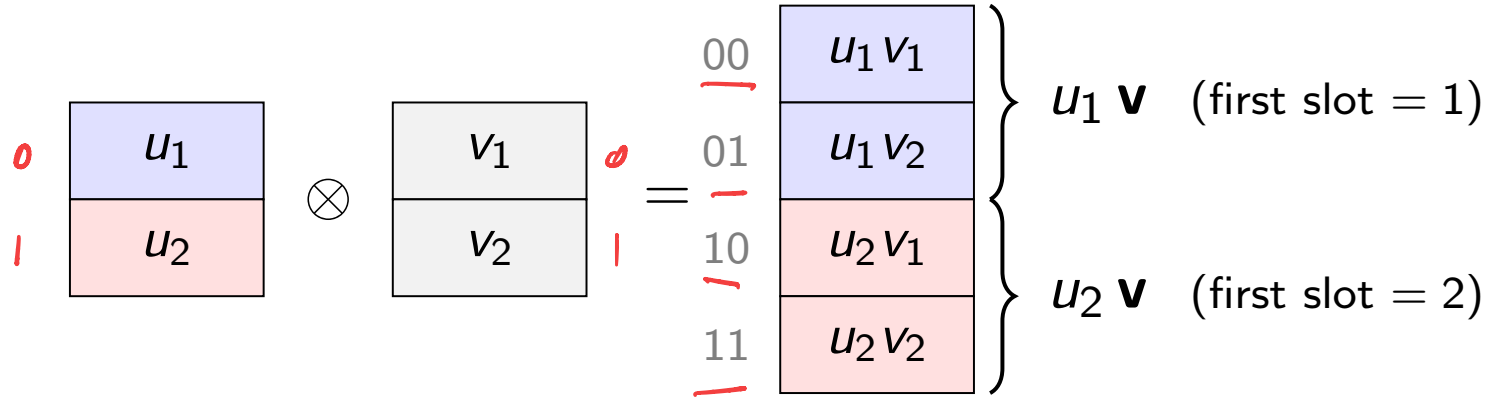
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Example

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ i \end{pmatrix} = \begin{pmatrix} 3 \\ i \\ 6 \\ 2i \end{pmatrix}, \qquad \begin{pmatrix} 3 \\ i \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ i \\ 2i \end{pmatrix}.$$

Order matters: $\mathbf{u} \otimes \mathbf{v} \neq \mathbf{v} \otimes \mathbf{u}$ *in general. "First qubit" and "second qubit" are different slots.*

Picture: Each Entry of $\mathbf{u}$ Scales a Copy of $\mathbf{v}$



- Entry (j, k) of $\mathbf{u} \otimes \mathbf{v}$ is $u_j v_k$: one index from each factor.
- Reading the positions as bit strings $00, 01, 10, 11$ (left bit from $\mathbf{u}$) gives the qubit labelling.

The Standard Basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and Kets

Tensor products of basis vectors give the standard basis of $\mathbb{C}^4$:

$$\mathbf{e}_1 \otimes \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_1 \otimes \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 \otimes \mathbf{e}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 \otimes \mathbf{e}_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

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In ket notation

With $|0\rangle = \mathbf{e}_1$, $|1\rangle = \mathbf{e}_2$ we write $|ab\rangle = |a\rangle \otimes |b\rangle$:

$$|00\rangle, \ |01\rangle, \ |10\rangle, \ |11\rangle$$

and a general two-qubit state is $\psi = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$, $\sum |\alpha_{ab}|^2 = 1$.

Also written $|a\rangle\,|b\rangle$ or $|a, b\rangle$. The tensor symbol is often dropped.

Rules for Computing

$|11\rangle \equiv |1\rangle \otimes |1\rangle$

For vectors $\mathbf{u}, \mathbf{u}', \mathbf{v}, \mathbf{v}'$ and scalars α:

Bilinearity

$$(\mathbf{u} + \mathbf{u}') \otimes \mathbf{v} = \mathbf{u} \otimes \mathbf{v} + \mathbf{u}' \otimes \mathbf{v}, \qquad \mathbf{u} \otimes (\mathbf{v} + \mathbf{v}') = \mathbf{u} \otimes \mathbf{v} + \mathbf{u} \otimes \mathbf{v}',$$

$$(\alpha \mathbf{u}) \otimes \mathbf{v} = \mathbf{u} \otimes (\alpha \mathbf{v}) = \alpha\,(\mathbf{u} \otimes \mathbf{v}).$$

$\begin{pmatrix}1\\2\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} \quad \begin{pmatrix}1\\3\end{pmatrix}$

$= \begin{pmatrix}1\\3\end{pmatrix} \otimes \begin{pmatrix}1\\3\end{pmatrix} = \begin{pmatrix}1\\3\\3\\9\end{pmatrix}$

$\begin{pmatrix}1\\2\end{pmatrix} \otimes \begin{pmatrix}1\\3\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}1\\3\end{pmatrix}$

$= \begin{pmatrix}1\\3\\2\\6\end{pmatrix} + \begin{pmatrix}0\\0\\1\\3\end{pmatrix} = \begin{pmatrix}1\\3\\3\\9\end{pmatrix}$

Rules for Computing

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$$(\alpha\mathbf{u})\otimes\mathbf{v} = \mathbf{u}\otimes(\alpha\mathbf{v}) = \alpha(\mathbf{u}\otimes\mathbf{v}).$$

Inner products multiply

$$\langle \mathbf{u}\otimes\mathbf{v},\, \mathbf{u}'\otimes\mathbf{v}'\rangle = \langle \mathbf{u},\, \mathbf{u}'\rangle \langle \mathbf{v},\, \mathbf{v}'\rangle, \qquad \|\mathbf{u}\otimes\mathbf{v}\| = \|\mathbf{u}\|\ \|\mathbf{v}\|.$$

Consequence: the tensor product of two unit vectors is a unit vector, and of two ONBs is an ONB.

$\langle u\otimes v,\ u\otimes v\rangle = \underbrace{\langle u, u\rangle}_{1} \cdot \underbrace{\langle v, v\rangle}_{1}$

Example: Expanding $|+\rangle \otimes |+\rangle$

Using bilinearity, like multiplying out brackets (but keeping the order):

$$|+\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \otimes \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) = \frac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big).$$

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As a column: $\frac{1}{2}(1,1,1,1)^T$, an equal superposition of all four bit strings.

$$|+\rangle \otimes |+\rangle \otimes |+\rangle = \frac{1}{2\sqrt{2}}\left(|000\rangle + |010\rangle + \ldots\right)$$

$|0\rangle \otimes |0\rangle \otimes |0\rangle$

Example: Expanding $|+\rangle \otimes |+\rangle$

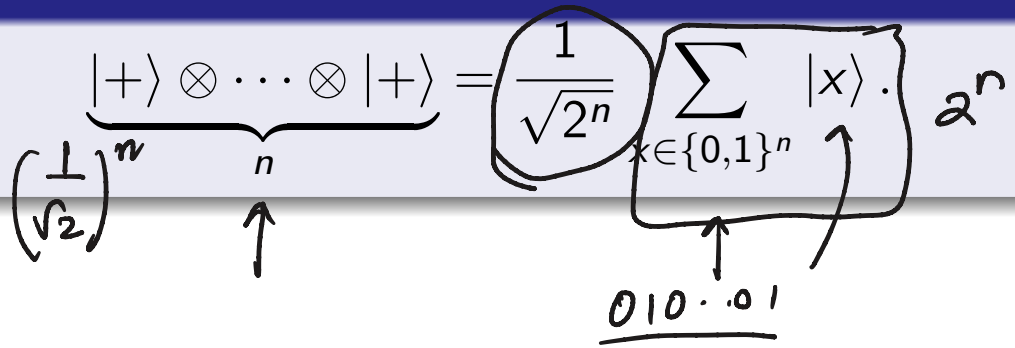
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n qubits

$$\underbrace{|+\rangle \otimes \cdots \otimes |+\rangle}_{n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle .$$



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Mixed example

$$|0\rangle \otimes |-\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle - |01\rangle\big) = \tfrac{1}{\sqrt{2}}(1,-1,0,0)^T.$$

Dimensions Multiply

$$\dim(\mathbb{C}^m \otimes \mathbb{C}^n) = mn$$

so n qubits live in $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$.

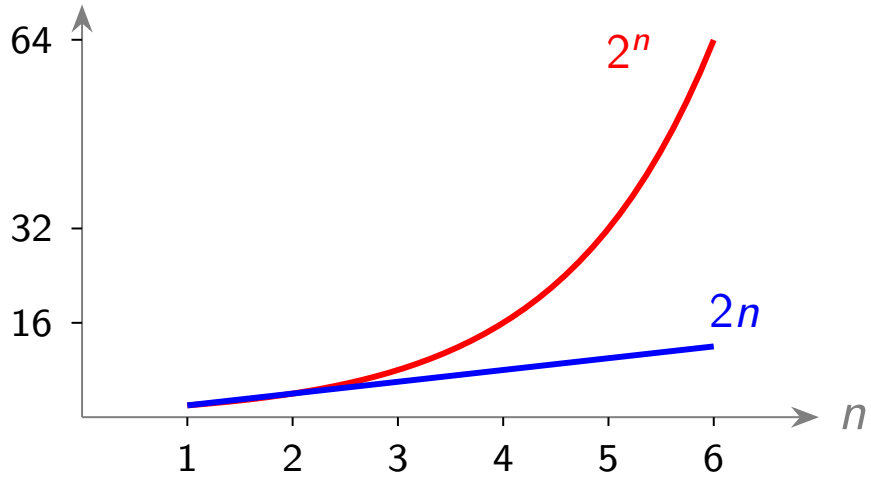
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qubits n	numbers for n separate qubits	amplitudes 2^n
2	4	4
10	20	1,024
30	60	$\approx 10^9$
→ 50	100	$\approx 10^{15}$

Describing n qubits separately needs 2n numbers; a general n-qubit state needs 2^n.



Which Vectors in $\mathbb{C}^4$ Are Tensor Products?

A two-qubit state is a **product state** if $\psi = \mathbf{u} \otimes \mathbf{v}$ for some $\mathbf{u}, \mathbf{v} \in \mathbb{C}^2$.

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = (\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha'|0\rangle + \beta'|1\rangle)$$

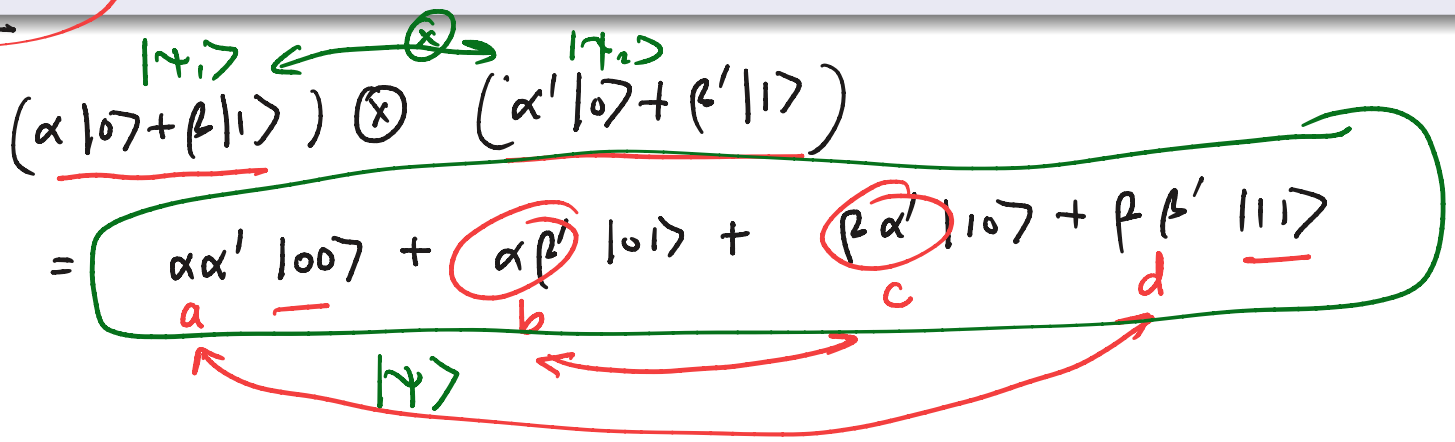
$|\psi_1\rangle$ $|\psi_2\rangle$

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A simple test

$\psi = (a, b, c, d)^T$ (with $\psi \neq \mathbf{0}$) is a product state $\iff$ $ad - bc = 0$.



$ad = bc$.

$a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$

$\stackrel{?}{=} (\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha'|0\rangle + \beta'|1\rangle)$

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Why ($\Rightarrow$): $(a, b, c, d) = (u_1 v_1, u_1 v_2, u_2 v_1, u_2 v_2)$, so $ad = u_1 v_1 u_2 v_2 = bc$.

Why ($\Leftarrow$): $ad = bc$ says the two halves (a, b) and (c, d) are parallel, i.e. both are multiples of one $\mathbf{v}$.

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Why ($\Leftarrow$): $ad = bc$ says the two halves (a, b) and (c, d) are parallel, i.e. both are multiples of one $\mathbf{v}$.

Example

$\frac{1}{2}(1, 1, 1, 1)$: $ad - bc = \frac{1}{4} - \frac{1}{4} = 0$, product $(|+\rangle\,|+\rangle)$. ✓

Arrange (a, b, c, d) as the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: product $\iff$ its determinant is 0.

Entangled States

Definition

A state of two (or more) qubits that is **not** a product state is called **entangled**.

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The Bell state

$\mathbf{\Phi}^+ = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(1,0,0,1)^T$: $ad - bc = \frac{1}{2} \neq 0$. Entangled.

Bell state

$\neq (\alpha|0\rangle + \beta|1\rangle) \otimes (\alpha'|0\rangle + \beta'|1\rangle)$

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The four Bell states (an ONB of $\mathbb{C}^4$)

$$\mathbf{\Phi}^{\pm} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \qquad \mathbf{\Psi}^{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

Entangled: there is no way to assign each qubit its own state. The pair has a definite joint state; the parts do not.

Example

Product or entangled? If product, factor it.

1. $\frac{1}{2}(1,\ 1,\ -1,\ -1)^T$
2. $\frac{1}{2}(1,\ 1,\ 1,\ -1)^T$
3. $\frac{1}{\sqrt{2}}(0,\ 1,\ i,\ 0)^T$
4. $\frac{1}{2}(1,\ i,\ i,\ -1)^T$

Use $ad - bc$ first; then read off $\mathbf{u}$ from how the two halves are scaled.

work here

Solutions

1. $ad - bc = -\frac{1}{4} + \frac{1}{4} = 0$: product. Halves $\frac{1}{2}(1,1)$ and $-\frac{1}{2}(1,1)$, so $\frac{1}{\sqrt{2}}(1,-1)^T \otimes \frac{1}{\sqrt{2}}(1,1)^T = |-\rangle \otimes |+\rangle$.
2. $ad - bc = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$: entangled.
3. $ad - bc = 0 - \frac{i}{2} \neq 0$: entangled.
4. $ad - bc = -\frac{1}{4} - \frac{i^2}{4} = 0$: product. Second half $= i \times$ first half, so $\frac{1}{\sqrt{2}}(1,i)^T \otimes \frac{1}{\sqrt{2}}(1,i)^T = \mathbf{c}_+ \otimes \mathbf{c}_+$.

(2) is $\frac{1}{\sqrt{2}}(|0\rangle |+\rangle + |1\rangle |-\rangle)$: a perfectly natural-looking state that is entangled.

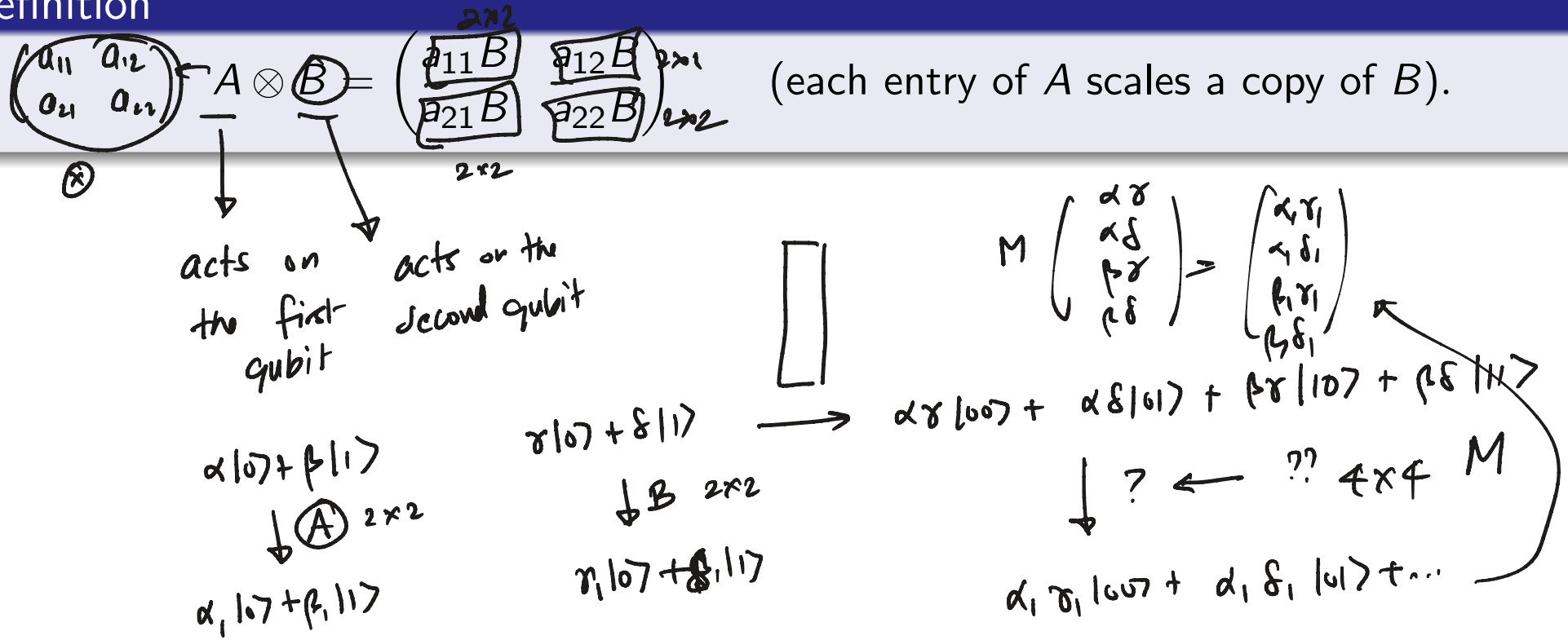
Tensor Product of Matrices

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \otimes \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \rightarrow \begin{pmatrix} u_1 v \\ u_2 v \end{pmatrix}$$

Definition

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix}$$

(each entry of A scales a copy of B).



Tensor Product of Matrices

Definition

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix} \qquad \text{(each entry of } A \text{ scales a copy of } B\text{)}.$$

$$X \otimes I = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} c \\ d \\ a \\ b \end{pmatrix}$$

flips the **first** qubit:

$|00\rangle \leftrightarrow |10\rangle, \quad |01\rangle \leftrightarrow |11\rangle$

$$I \otimes X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

flips the **second** qubit:

$|00\rangle \leftrightarrow |01\rangle, \quad |10\rangle \leftrightarrow |11\rangle$

$= c|00\rangle + d|01\rangle + a|10\rangle + b|11\rangle$

$(X \otimes I)(a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle)$

$= a|10\rangle + b|11\rangle + c|00\rangle + d|01\rangle$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{bmatrix} 4\times 4 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$M = A \otimes B$

$$X \otimes I = \begin{pmatrix} \boxed{0 \cdot I}^{\,2} & \boxed{1 \cdot I} \\ 1\,I & 0 \cdot I \end{pmatrix}^{2}$$

$$= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$

$\alpha|0\rangle + \beta|1\rangle \quad \gamma|0\rangle + \delta|1\rangle$

The Key Rule: Act on Each Factor Separately

Mixed-product rule

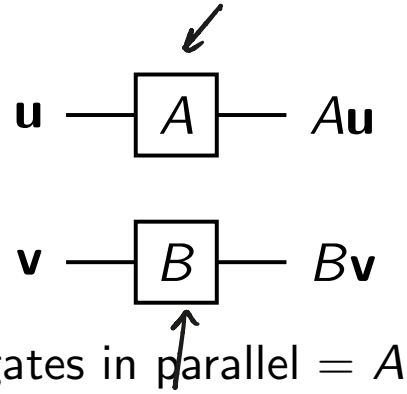
$$\left.\begin{aligned}(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) &= (A\mathbf{u}) \otimes (B\mathbf{v}) \\ (A \otimes B)(C \otimes D) &= (AC) \otimes (BD)\end{aligned}\right\}$$

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$$(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = (A\mathbf{u}) \otimes (B\mathbf{v})$$

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$



Two gates in parallel $= A \otimes B$

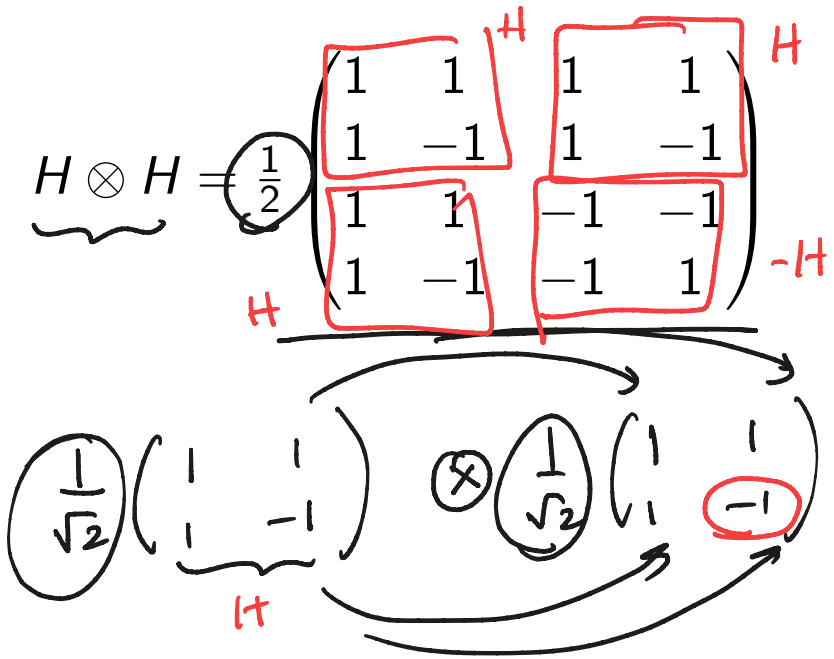
Consequences:

- $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$
- A, B unitary $\Rightarrow A \otimes B$ unitary
- $A \otimes B = (A \otimes I)(I \otimes B) = (I \otimes B)(A \otimes I)$

Gates on different qubits commute; the order in which you draw them in a single column does not matter.

$= A \cdot I \otimes I \cdot B = A \otimes B$

Example: Hadamard on Every Qubit



$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$\alpha|0\rangle \to \alpha \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$+$

$\beta|1\rangle \to \beta \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$\frac{1}{\alpha}$

Example: Hadamard on Every Qubit

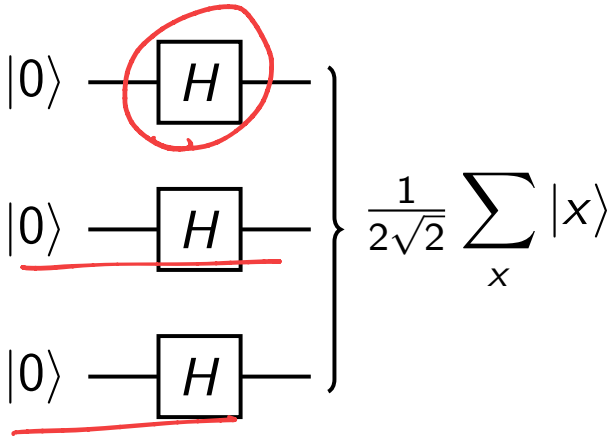
$$H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

By the mixed-product rule, no 4×4 multiplication is needed:

$$(H \otimes H)\,|00\rangle = H\,|0\rangle \otimes H\,|0\rangle = |+\rangle\,|+\rangle\,.$$

$H \otimes H \otimes H \;\; |000\rangle \;\; = \;\; |+\rangle|+\rangle|+\rangle$

$\frac{1}{2\sqrt{2}}\left(|000\rangle + |010\rangle + \cdots |111\rangle \right)$



$n = 3$: all 8 strings

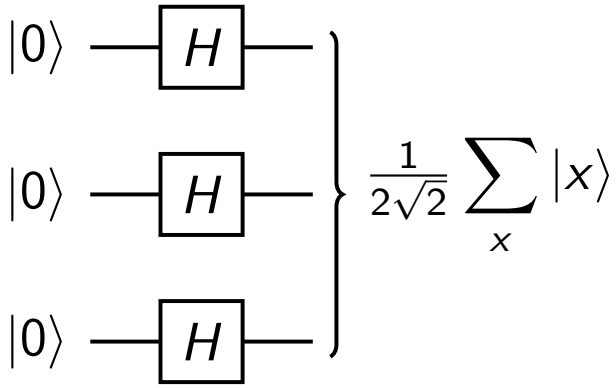
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n qubits

$H^{\otimes n}\,|0 \cdots 0\rangle = \dfrac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$: n gates create a superposition over 2^n strings.

This is the first step of almost every quantum algorithm.

Not Every Two-Qubit Gate Is a Tensor Product: CNOT

CNOT flips the second qubit iff the first is 1:

$$|a\rangle\,|b\rangle \mapsto |a\rangle\,|b \oplus a\rangle$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = P_0 \otimes I + P_1 \otimes X$$

$(P_0 = \mathbf{e}_1\mathbf{e}_1^\dagger,\ P_1 = \mathbf{e}_2\mathbf{e}_2^\dagger).$

$= b$ if $a = 0$

$= \bar{b}$ if $a = 1$

$0 \oplus 0 = 0$

$0 \oplus 1 = 1$

$1 \oplus 1 = 0$

$1 \oplus 0 = 1$

$b \oplus 1 =$ neg. of b.

$b \oplus 0 = b$

$|101\rangle \to |101\rangle$

$\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$

$\alpha|00\rangle + \beta|01\rangle + \gamma|11\rangle + \delta|10\rangle$

$\neq A \otimes B$ (2×2, 2×2)

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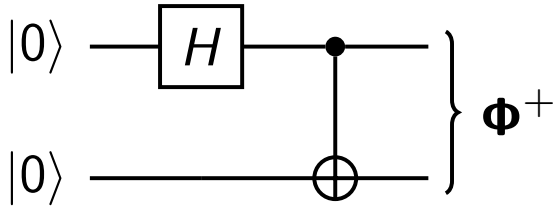
$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Creating entanglement

$$|00\rangle \xrightarrow{H\otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

a b a b

"entayled"



Product gates $A \otimes B$ keep product states product. Creating entanglement needs a gate like CNOT.

Application 1: Measuring Only the First Qubit

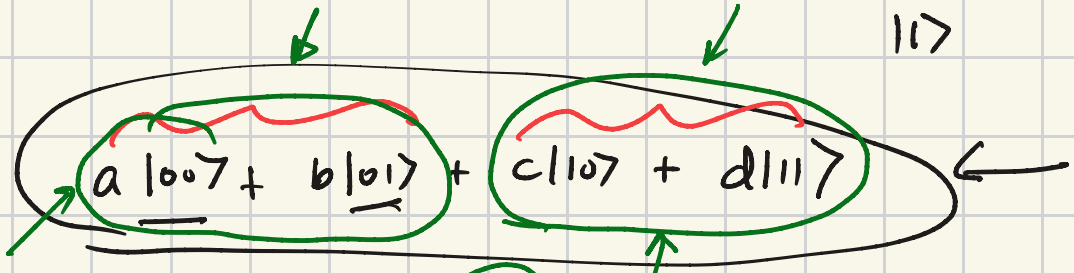
The projectors for "first qubit is 0 / 1" are $P_0 \otimes I$ and $P_1 \otimes I$, with $P_0 = \mathbf{e}_1\mathbf{e}_1^\dagger$, $P_1 = \mathbf{e}_2\mathbf{e}_2^\dagger$.

Rule

$\Pr(\text{first} = a) = \|(P_a \otimes I)\psi\|^2$; afterwards the state is $(P_a \otimes I)\psi$, renormalised.

what happens when we measure qubits?

$\alpha|0\rangle + \beta|1\rangle$?



$(|0\rangle \otimes (a|0\rangle + b|1\rangle))$

$= \dfrac{a|00\rangle + b|01\rangle}{\sqrt{|a|^2+|b|^2}}$

$\begin{cases} |00\rangle \\ |01\rangle \end{cases}$ $|a|^2$, $|b|^2$ ← prob.

$|10\rangle$: $|c|^2$

$|11\rangle$: $|d|^2$

measure both qubits.

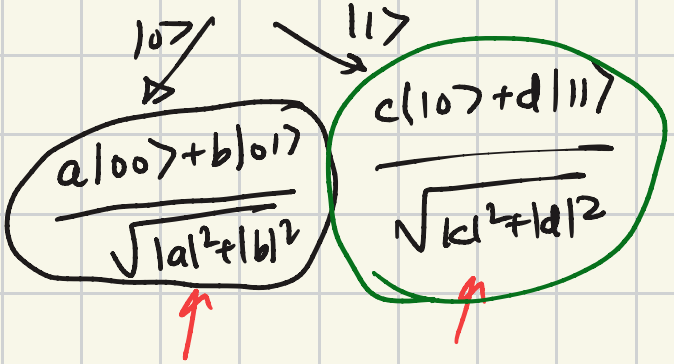
Pr. first qubit is $|0\rangle$ $|a|^2+|b|^2$

Pr. first $|1\rangle$ $|c|^2+|d|^2$

$|0\rangle$ $|a|^2$

$|1\rangle$ $|\beta|^2$

measure the first qubit?



what is the prob. that the first qubit is $|0\rangle$? → $|a|^2+|b|^2$

$|c|^2+|d|^2$

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Example

$\psi = \frac{1}{\sqrt{3}}\big(|00\rangle + |01\rangle + |10\rangle\big)$.

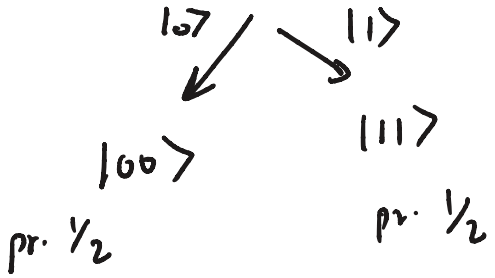
- $(P_0 \otimes I)\psi = \frac{1}{\sqrt{3}}\big(|00\rangle + |01\rangle\big)$: probability $\frac{2}{3}$, new state $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes |+\rangle$.
- $(P_1 \otimes I)\psi = \frac{1}{\sqrt{3}}\,|10\rangle$: probability $\frac{1}{3}$, new state $|10\rangle$.

The tensor product tells us exactly how "do nothing to the second qubit" is written: $\otimes I$.

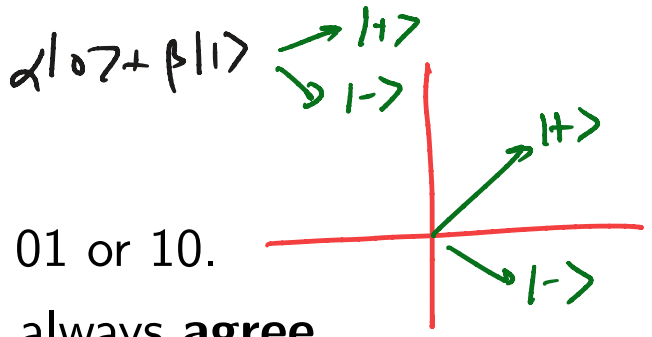
Application 2: Bell-State Correlations

Measure both qubits of $\mathbf{\Phi}^+ = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

- outcomes 00 or 11, each with probability $\frac{1}{2}$; never 01 or 10.
- Each qubit alone looks like a fair coin, but the two always **agree**.



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Same state, different basis (bilinearity)

$$\frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle) = \frac{1}{2\sqrt{2}}\Big[(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)(|0\rangle - |1\rangle)\Big] = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

$|+\rangle \quad |-\rangle$

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So measuring both qubits in the $\pm$ basis *also* gives perfectly agreeing outcomes.

A classical "both coins were secretly the same" explanation works for one basis, but cannot reproduce agreement in every basis at once. That gap is what Bell inequalities quantify.

Summary

- $\mathbf{u} \otimes \mathbf{v}$: each u_j scales a copy of $\mathbf{v}$; $|ab\rangle = |a\rangle \otimes |b\rangle$
- Bilinear; inner products and norms multiply
- n qubits live in $\mathbb{C}^{2^n}$
- Product iff $ad - bc = 0$ (two qubits); otherwise entangled
- $(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$
- $H^{\otimes n} |0^n\rangle =$ uniform superposition
- CNOT creates entanglement; $\otimes I$ means "leave alone"

Take away

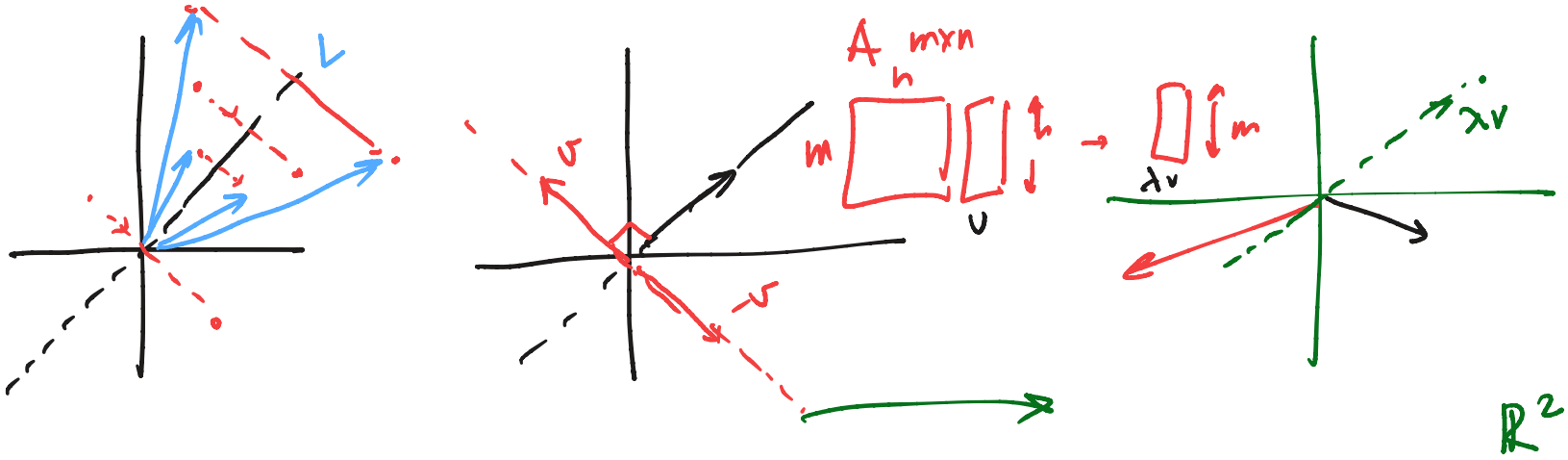
The tensor product is how quantum systems combine. Its size (2^n) and its non-product states (entanglement) are what make quantum computation different.

Eigenvalues and Eigenvectors

Definition

A non-zero vector $\mathbf{v}$ is an **eigenvector** of the square matrix A with **eigenvalue** $\lambda \in \mathbb{C}$ if

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4b

$$S = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$e_1 \to e_2 \to e_3$

$e_2 \to e_3 \to e_1$

$e_3 \to e_1 \to e_2$

$e_1 \to e_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$$S^2 = \begin{bmatrix} \quad \\ S \end{bmatrix} \cdot \begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

S^2

$e_1 \to e_2 + e_3$

$e_2 \to e_1 + e_2 + e_3$

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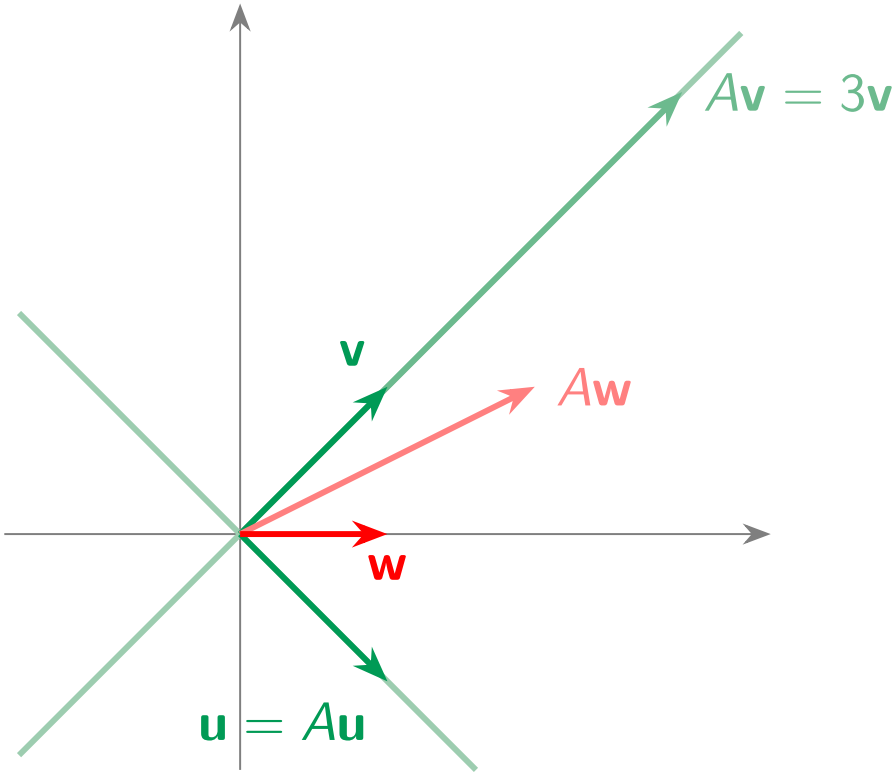
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"Eigen" is German for "own" or "characteristic": the directions that belong to A itself.

Picture: Directions That Survive

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

- $A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: stays on its line, stretched by 3.
- $A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$: knocked off its line.
- $A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$: eigenvalue 1.



Eigenvectors by Inspection

Some we already know, from how the gates act:

Matrix	Eigenvector	Eigenvalue	Why
Z	$\mathbf{e}_1$	$+1$	$Z\mathbf{e}_1 = \mathbf{e}_1$
	$\mathbf{e}_2$	-1	$Z\mathbf{e}_2 = -\mathbf{e}_2$
X	$\mathbf{h}_+$	$+1$	X swaps the entries of $(1,1)^T$
	$\mathbf{h}_-$	-1	$X(1,-1)^T = (-1,1)^T$
I	every $\mathbf{v} \neq \mathbf{0}$	1	$I\mathbf{v} = \mathbf{v}$
$\mathrm{diag}(d_1, \ldots, d_n)$	$\mathbf{e}_j$	d_j	scales each axis

Diagonal matrices are the easy case: the eigenvectors are the standard basis. Much of today is about making a matrix look diagonal.

Eigenvectors Are Not Unique: Eigenspaces

- If $A\mathbf{v} = \lambda\mathbf{v}$ then $A(\alpha\mathbf{v}) = \lambda(\alpha\mathbf{v})$ for any $\alpha \neq 0$, including complex α.
- If $\mathbf{v}, \mathbf{w}$ both have eigenvalue λ, so does $\mathbf{v} + \mathbf{w}$.

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Convention

We usually pick a **unit** eigenvector. It is still only fixed up to a phase $e^{i\varphi}$, and that phase never matters.

Example: for I, $E_1 = \mathbb{C}^n$, the whole space.

Finding Eigenvalues: The Characteristic Polynomial

$$A\mathbf{v} = \lambda\mathbf{v},\ \mathbf{v} \neq \mathbf{0} \iff (A - \lambda I)\mathbf{v} = \mathbf{0} \text{ has a non-zero solution} \iff \det(A - \lambda I) = 0.$$

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Recipe

1. Solve $\det(A - \lambda I) = 0$: a degree-n polynomial in λ.
2. For each root λ, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for $\mathbf{v}$.

Worked Example: Y

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Why Complex Numbers Again: Rotations

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ rotates $\mathbb{R}^2$ by θ.

For $0 < \theta < \pi$ **no real direction** survives.

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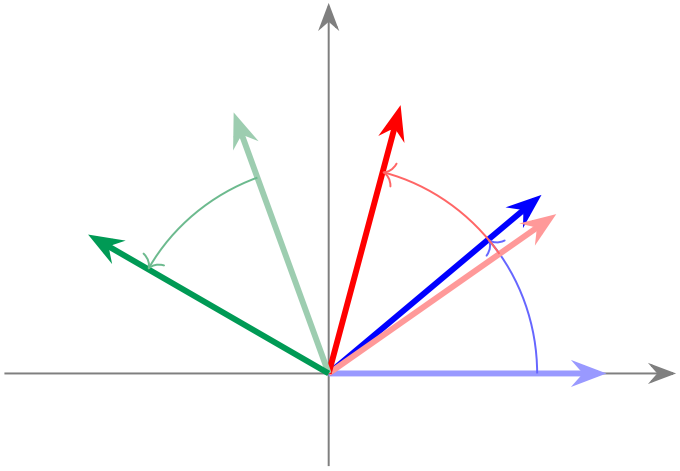
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Eigenvectors: $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$ for $e^{i\theta}$, $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$ for $e^{-i\theta}$.

Complex eigenvalues on the unit circle: a preview of unitary matrices.



every arrow turns:
no real eigenvector

Diagonalisation

Suppose A has n independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$. Put them as columns of V:

$$AV = A\big(\mathbf{v}_1 | \cdots | \mathbf{v}_n\big) = \big(\lambda_1 \mathbf{v}_1 | \cdots | \lambda_n \mathbf{v}_n\big) = VD, \qquad D = \mathsf{diag}(\lambda_1, \ldots, \lambda_n).$$

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In the eigenbasis, A just scales each coordinate: *A is diagonal in its own eigenbasis.*

Part 3 upgrade: when the eigenvectors are orthonormal, V is unitary and $V^{-1} = V^{\dagger}$. No inverse needed.

Two Useful Facts

If $A\mathbf{v} = \lambda\mathbf{v}$ then:

Powers

$A^2\mathbf{v} = A(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$, and in general $A^k\mathbf{v} = \lambda^k\mathbf{v}$.
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Polynomials

For $p(t) = a_0 + a_1 t + \cdots + a_m t^m$: $\quad p(A)\mathbf{v} = p(\lambda)\mathbf{v}$.

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Consequence

$X^2 = I$, so if $X\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v} = \lambda^2\mathbf{v}$ and $\lambda = \pm 1$. The same argument works for Y, Z, H: *any matrix with $M^2 = I$ has eigenvalues ± 1.*

Keep "$p(A)$ acts as $p(\lambda)$ on eigenvectors" in mind; it is the whole idea of Part 4.

Examples

1. Eigenvalues and eigenvectors of $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.
2. Eigenvalues of $S = \text{diag}(1, i)$ and $T = \text{diag}(1, e^{i\pi/4})$.
3. Find an eigenvector of H with eigenvalue $+1$.
4. Show: if A is invertible, no eigenvalue is 0.

work here

Solutions

1. $\lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$. $\lambda = 3$: $\mathbf{h}_+$; $\lambda = -1$: $\mathbf{h}_-$.
2. Diagonal, so read them off: S: $1, i$; T: $1, e^{i\pi/4}$. Eigenvectors $\mathbf{e}_1, \mathbf{e}_2$.
3. $(H - I)\mathbf{v} = \mathbf{0}$: $(\frac{1}{\sqrt{2}} - 1)v_1 + \frac{1}{\sqrt{2}}v_2 = 0 \Rightarrow v_2 = (\sqrt{2} - 1)v_1$. So $\mathbf{v} = (1, \sqrt{2} - 1)^T$ (normalise if you like).
4. If $A\mathbf{v} = 0 \cdot \mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{v} = A^{-1}\mathbf{0} = \mathbf{0}$. Contradiction.

In (3) the eigenvector points at angle $\pi/8$: H is a reflection across that line.

Summary: Eigenvalues and Eigenvectors

- $A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$
- Eigenvalues: roots of $\det(A - \lambda I)$
- 2×2: $\lambda^2 - (\operatorname{Tr} A)\lambda + \det A = 0$
- Over $\mathbb{C}$, eigenvalues always exist
- Diagonalisable: $A = VDV^{-1}$
- $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$

Take away

Eigenvectors are the directions a matrix only stretches. In that basis the matrix is diagonal, and everything becomes easy.